OPERATION RESEARCH CHAPTER 3 MCQs

1. Linear programming is primarily concerned with:

A. Solving only nonlinear equations

B. Allocation of limited resources among competing activities

C. Finding roots of algebraic equations

D. Forecasting future demand only

2. Which condition strictly violates the proportionality assumption in a linear programming model?

 A. Incurring an initial fixed setup cost when initiating an activity

 B. Having simultaneous non-zero production quantities across activities

 C. Allowing decision variables to take continuous fractional numbers

 D. Setting a strict upper bound capacity on total resource consumption

3. Under what circumstance is the additivity assumption of linear programming violated?

 A. When decision variables are restricted to integer values

 B. When an objective function is minimized rather than maximized

 C. When synergy between two activities yields extra interaction profit

 D. When the value assigned to a constraint limit varies over time

4. What is the relationship between optimal solutions and corner-point feasible (CPF) solutions in a bounded, feasible linear programming problem?

 A. All CPF solutions must yield identical objective values

 B. An optimal solution is guaranteed to lie in the interior

 C. At least one optimal solution must be a CPF solution

 D. Every feasible solution must also be a CPF solution

5. What does the divisibility assumption in linear programming require?

 A. Decision variables are allowed to take any non-integer real values

 B. All functional constraints must be expressed as exact equalities

 C. Objective coefficients can be partitioned across different time periods

 D. Each plant capacity can be divided equally across every product

6. The principal purpose of an LP model is to:

A. Find all possible solutions

B. Find the best feasible allocation of resources

C. Eliminate all constraints

D. Make all decision variables equal

7. Which condition can lead to a linear programming problem having no optimal solution?

 A. Having an objective function that can improve without bound

 B. Having more functional constraints than decision variables

 C. Having a non-zero constant in the objective function definition

 D. Requiring some functional constraints to hold as strict equations

8. How does standard linear programming handle the certainty assumption when real-world parameters contain predictive errors?

 A. By rounding all parameters up to conservative integer figures

 B. By performing sensitivity analysis after solving the model

 C. By replacing all inequality constraints with strict equalities

 D. By forcing the simplex method to evaluate only interior points

9. In a linear programming model, what defines a 'functional constraint'?

 A. The single mathematical equation that establishes the performance goal

 B. A software macro computing summations over data indices in solvers

 C. A restriction expressing resource usage as a linear function of variables

 D. A nonnegativity restriction ensuring variables remain positive or zero

10. What is the primary operational advantage of using mathematical modelling languages such as MPL or LINGO?

 A. They enable compact index-based formulations of very large models

 B. They bypass the need to solve problems using the simplex method

 C. They guarantee that non-convex problems find global optima instantly

 D. They convert nonlinear cross-product terms directly into linear terms

11. Parallel objective-function lines correspond to:

A. Different objective-function coefficients

B. Different values of the same objective function

C. Different feasible regions

D. Different constraints

12. In the graphical method, the optimal solution generally occurs:

A. Outside the feasible region

B. At a corner point of the feasible region

C. At the origin only

D. At the midpoint of every constraint

13. If Z increases while objective coefficients remain unchanged, the objective line:

A. Rotates

B. Remains parallel and shifts

C. Becomes curved

D. Disappears

14. The general maximization objective is:

A. $Z = b1 + ... + bm$

B. $Z = c1x1 + c2x2 + ... + cnxn$

C. $Z = a11 + a22$

D. $Z = x1x2$

15. The conditions $xi >= 0$ are known as:

A. Binding constraints

B. Resource constraints

C. Equality constraints

D. Nonnegativity constraints

16. What is an Infeasible solution

A. Violates at least one constraint

B. Maximizes the objective function

C. Is always at the origin

D. Is necessarily negative

17. An optimal solution is:

A. Any feasible solution

B. An infeasible solution with high profit

C. A feasible solution having the most favourable objective value

D. A solution satisfying one constraint

18. A corner-point feasible solution is located:

A. Outside the feasible region

B. At a corner of the feasible region

C. At the origin only

D. On the objective axis only

19. If an LP problem has exactly one optimal solution, then:

A. It must be a corner-point feasible solution

B. It cannot be a corner point

C. It must lie outside the feasible region

D. It must have negative variable

20. If no solution satisfies all constraints, the LP problem has:

A. Multiple optima

B. No feasible solution

C. An unbounded optimum

D. A unique optimum

21. If the objective value can increase indefinitely in a maximization problem, the problem has:

A. No feasible solution

B. An unbounded solution

C. Multiple bounded solutions only

D. A degenerate solution necessarily

22. Which is NOT one of the basic assumptions of linear programming discussed in the chapter?

A. Proportionality

B. Additivity

C. Divisibility

D. Randomness

23. The proportionality assumption applies to:

A. Objective function only

B. Constraints only

C. Both objective function and functional constraints

D. Nonnegativity only

24.Under proportionality, doubling an activity level causes its contribution to the objective function to:

A. Stay unchanged

B. Double

C. Become zero

D. Quadruple

25. A fixed start-up cost is most likely to violate:

A. Proportionality

B. Certainty

C. Divisibility

D. Nonnegativity

26. The additivity assumption requires:

A. The total contribution to equal the sum of individual contributions

B. Every variable to be an integer

C. All parameters to be uncertain

D. All constraints to be equality constraints

27. The divisibility assumption means:

A. Decision variables must be even numbers

B. Decision variables can take fractional values

C. Decision variables must be integers

D. Decision variables must be binary

28. If an activity can only be undertaken in whole units, the divisibility assumption may:

A. Hold perfectly

B. Be violated

C. Be irrelevant in every case

D. Imply infeasibility

29. When fractional decision-variable values are unacceptable, a more suitable model may be:

A. Integer programming

B. Pure graphical programming

C. Regression analysis

D. Queuing theory

30. The certainty assumption states that:

A. All decision variables are known beforehand

B. Every model parameter is assumed to be a known constant

C. All resources are unlimited

D. The optimum is predetermined

31- Sensitivity analysis is particularly useful when:

A. Model parameters involve uncertainty

B. There are no decision variables

C. All constraints are removed

D. The model has no objective

32. Solve graphically:

Maximize Z = 2x1 + 3x2

subject to

x1 <= 10

2x1 + 5x2 <= 60

x1 + x2 <= 18

3x1 + x2 <= 44

x1, x2 >= 0

Find all corner points, optimal x1, x2, and maximum Z.

33. Solve graphically:

Maximize Z = 10x1 + 20x2

subject to

x1 + 2x2 <= 15

x1 + x2 <= 12

5x1 + 3x2 <= 45

x1, x2 >= 0

Find the optimal solution and maximum objective value

34. Solve graphically:

Maximize Z = 15x1 + 20x2

subject to

x1 + 2x2 >= 10

2x1 - 3x2 <= 6

x1 + x2 <= 6 x1, x2 >= 0

Determine the feasible region and optimal solution

35. Solve graphically:

Minimize Z = 3x1 + 2x2

subject to x1 + 2x2 >= 12

2x1 + 3x2 >= 12

2x1 + x2 >= 8

x1, x2 >= 0

36. Maximize Z = c1x1 + 2x2

subject to 4x1 + x2 <= 12

x1 + 2x2 <= 8

x1, x2 >= 0

Determine the range of c1 for which a given corner point remains optimal

37. Determine whether the following problem is unbounded:

Maximize Z = 2x1 + x2

subject to

x1 - x2 <= 10,

x1 + x2 >= 30,

x1,x2 >= 0.

38. MARY has just been diagnosed as having a cancer at a fairly advanced Stage. Specifically, she has a large malignant tumour in the bladder area (a "whole bladder lesion"). Mary is to

receive the most advanced medical care available to give her every possible chance for survival. This care will include extensive radiation therapy. Radiation therapy involves using an external beam treatment machine to pass ionizing radiation through the patient's body, damaging both cancerous and healthy tissues. Normally, several beams are precisely administered from different angles in a two- dimensional plane. Due to attenuation, each beam delivers more radiation to the tissue near the entry than to the tissue near the exit point. Scatter also causes some delivery of radiation to tissue outside the direct path of the beam. Because tumour cells are typically microscopically interspersed among healthy cells, the radiation dosage throughout the tumour region must be large enough to kill the malignant cells, which are slightly more radiosensitive, yet small enough to spare the healthy cells. At the same time, the aggregate dose to critical tissues must not exceed established tolerance levels, in order to pre- vent complications that can be more serious than the discase itself. For the same reason, the total dose to the entire healthy anatomy must be minimized. Because Of the need to carefully balance all these factors, the design of radiation therapy is a very delicate process The goal of the design is to select the combination of beams to be used, and the intensity of each one, to generate the best possible dose distribution. (The dose strength at any point in the body is measured in units called kilorads.) Once the treatment design been developed, it is administered in many installments, spread over several weeks. In Mary's case, the and location of her tumour make the design of her treatment an even more delicate process than usual. Figure 3. I I shows a diagram of a cross section of the tumour viewed from above, as well as nearby critical tissues to avoid. These tissues include critical organs (e.g., the rectum) as well as bony structures (e.g., the femurs and pelvis) that will attenuate the radiation. Also shown are the entry point and direction for the only two beams that can be used with any modicum of safety in this case. For any proposed beam of given intensity, the analysis of what the resulting radiation absorption by various parts of the body would be requires a complicated process. In brief, based on careful anatomical analysis, the energy distribution within the two- dimensional cross section of the tissue can be plotted on an isodose map, where the con tour lines represent the dose strength as a percentage of the dose strength at the entry point. A fine grid then is placed over the isodose map. By summing the radiation absorbed in the squares containing each type of tissue, the average dose that is absorbed by the tumour, healthy anatomy, and critical tissues can be calculated. With more than one beam (administered sequentially), the radiation absorption is additive. After thorough analysis of this type, the medical team has carefully estimated the data needed to design Mary 's treatment, as summarized in Table 3 The first column lists the areas of the body that must be considered, and then the next two columns give the frac- Lion Of the radiation dose at the entry point for each beam that is absorbed by the respective areas on average. For example, if the dose level at the entry point for beam I is I kilorad, then an average of 0.4 kilorad Will be absorbed by the entire healthy anatomy in the two-dimensional plane, an average of 0.3 kilorad will be absorbed by nearby crit- iCal tissues, an average of 0.5 kilorad will be absorbed by the various parts of the tumour, and 0.6 kilorad will be absorbed by the centre of the tumour. The last column gives the re- strictions on the total dosage from both beams that is absorbed on average by the respective areas of the body. In particular, the average dosage absorption for the healthy anatomy must be as small as possible, the critical tissues must not exceed 2.7 kilorads, the average over the entire tumour must equal 6 kilorads, and the centre of the tumour must be at least 6 kilorads.

Based on above given data, formulate the linear programming problem and solve it by simplex method or Big M method.

39. The agricultural output of each kibbutz is limited by both the amount of available irrigable land and the quantity of water allocated for irrigation by the Water Commissioner (a national government official). data are given

Kibbutz	*Usable Land (acres)*	*Water Allocation (acre Feet)*
1	*400*	*600*
2	*600*	*800*
3	*300*	*375*

The crops suited for this region include sugar beets, cotton, and sorghum, and these are the three being considered for the upcoming season. These crops differ primarily in their expected net return per acre and their consumption of water. In addition, the Ministry of Agriculture has set a maximum quota for the total acreage that can be devoted to each of these crops by the Southern Confederation of Kibbutzim are Given,

Crop	Maximum Quota (acres)	Water Consumption (acre feet/ acre)	Net Return($/acre)
Sugar Beets	600	3	1000
Cotton	500	2	750
Sorghum	325	1	250

Because of the limited water available for irrigation, the Southern Confederation of Kibbutzim will not be able to use all its irrigable land for planting crops in the upcoming season. To ensure equity between the three kibbutzim, it has been agreed that every kibbutz will plant the same proportion of its available irrigable land. For example, if kibbutz I plants 200 Of its available 400 acres, then kibbutz 2 must plant 300 Of its 600 acres, while kibbutz 3 plants 150 acres of its 300 acres. However, any combination of the crops may be grown at any of the kibbutzim. The job facing the Coordinating Technical Office is to plan how many acres to devote to each crop at the respective kibbutzim while satisfying the given restrictions. The objective is to maximize the total net return to the South ern Confederation of Kibbutzim as a whole.

Formulate a liner programming problem from the above given data

40. The WYNDOR GIASS CO. produces high-quality glass products, including windows and glass doors. It has three plants. Aluminium frames and hardware are made in Plant I , wood frames are made in Plant 2, and Plant 3 produces the glass and assembles the products. Because Of declining earnings, top management has decided to revamp the company's product line. Unprofitable products are being discontinued, releasing production capacity to launch two new products having large sales potential: Product l: An 8-foot glass door with aluminium framing product 2: A 4 X 6-foot double-hung Wood framed window Product I requires some of the production capacity in Plants I and 3, but none in Plant 2. Product 2 needs only Plants 2 and 3. The marketing division has concluded that the company could sell as much of either product as could be produced by these plants. HOW ever, because both products would be competing for the same production capacity in Plant 3, it is not clear which mix of the two products would be most profitable. Therefore, an OR team has been formed to study this question. The OR team began by having discussions with upper management to identify management's objectives for the study. These discussions led to developing the following definition of the problem: Determine what the production rates should be for the two products in order to maximize their profit?

Formulate the linear programming problems.

41. The NORI & LEETS CO., one of the major producers of steel in its part of the world, is located in the city of Steeltown and is the only large employer there. Steeltown has grown and along with the company, which now employs nearly 50,000 residents. There- fore, the altitude of the townspeople always has been, "What's good Nori & Leets is good for the town." However, this attitude is now changing; uncontrolled air pollution from the company's furnaces is ruining the appearance of the city and endangering the health of its residents.

TABLE 3.12 Clean air standards for the Nori & Leets Co.

Pollutant	Required Reduction in Annual Emission Rate (Million Pounds)
Particulates	60
Sulfur oxides	150
Hydrocarbons	125

TABLE 3.13 Reduction in emission rate (in millions of pounds per year) from the maximum feasible use of an abatement method for Nori & Leets Co.

	Taller Smokestacks		Filters		Better Fuels	
Pollutant	Blast Furnaces	Open-Hearth Furnaces	Blast Furnaces	Open-Hearth Furnaces	Blast Furnaces	Open-Hearth Furnaces
Particulates	12	9	25	20	17	13
Sulfur oxides	35	42	18	31	56	49
Hydrocarbons	37	53	28	24	29	20

TABLE 3.14 Total annual cost from the maximum feasible use of an abatement method for Nori & Leets Co. ($ millions)

Abatement Method	Blast Furnaces	Open-Hearth Furnaces
Taller smokestacks	8	10
Filters	7	6
Better fuels	11	9

Formulate Linear Programming Model from the above given data.

42. UNION AIRWAYS is adding more flights to and from its hub airport, and so it needs to hire additional customer service agents. However, it is not clear just how many more should be hired. Management recognizes the need for cost control while also consistently providing a satisfactory level of service to customers. Therefore, an OR team is studying how to schedule the agents to provide satisfactory service with the smallest personnel cost. Based on the new schedule of nights, an analysis has been made of the minimum number of customer service agents that need to be on duty at different times of the day to provide a satisfactory level of service. The rightmost column Of Table 3.19 shows the number of agents needed for the time periods given in the first column. The other entries in this table reflect one of the provisions in the company's current contract with the union that represents the customer service agents. The provision is that each agent works an 8-hour shift 5 days per week, and the authorized shifts are

TABLE 3.19 Data for the Union Airways personnel scheduling problem

Time Period	Time Periods Covered: Shift 1	2	3	4	5	Minimum Number of Agents Needed
6:00 A.M. to 8:00 A.M.	✔					48
8:00 A.M. to 10:00 A.M.	✔	✔				79
10:00 A.M. to noon	✔	✔				65
Noon to 2:00 P.M.	✔	✔	✔			87
2:00 P.M. to 4:00 P.M.		✔	✔			64
4:00 P.M. to 6:00 P.M.			✔	✔		73
6:00 P.M. to 8:00 P.M.			✔	✔		82
8:00 P.M. to 10:00 P.M.				✔		43
10:00 P.M. to midnight				✔	✔	52
Midnight to 6:00 A.M.					✔	15
Daily cost per agent	$170	$160	$175	$180	$195	

Shift 1: 06:00 am – 02:00 pm

Shift 2: 02:00am – 04:00 pm

Shift 3: Noon- 08:00 pm

Shift 4: 04: pm- Midnight

Shift 5: 10:00 P.M. to 6:00 A.M.

Formulate Linear Programming Model based on above data.

Answers- 1-B, 2-A, 3-C, 4-C, 5-A, 6-B, 7-A, 8-B, 9-C, 10-A, 11-B, 12- B, 13- C, 14- B, 15- D, 16-A, 17- C, 18- B, 19- A, 20- B, 21- B, 22- D,23- C, 24- B, 25- A, 26- A, 27- B, 28- B, 29-A, 30- B, 31- A,